

Concept 1 | Laws of Exponents and Roots

English Student Edition • Focused formulas and guided practice

Formula opener

Product

$$a^m a^n = a^{m+n}$$

Power

$$(a^m)^n = a^{mn}$$

Root

$$a^{1/n} = \sqrt[n]{a}$$

Classified Practice

Laws of Exponents and Roots

Q001 Written

Find: (a) 10^0 , (b) 2^{-3} , (c) $(0.2)^{-2}$.

Q002 Written

Let $a \neq 0, b \neq 0$. Simplify: (a) $a^{-5}a^8$, (b) $a^5 \div a^{-3}$, (c) $(a^{-1}b)^2$, (d) $(2^2)^3 \div 2^{-2} \times 2^{-3}$.

Classified Practice

Laws of Exponents and Roots

Q003 Written

Find: (a) 5^0 , (b) 3^{-2} , (c) $(0.25)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q004 Written

Simplify the eight given expressions: a^5a^4 , $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, **and** $2^4 \div (2^{-1})^3 \times 2^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q005 Written

Find the nine values: 3^0 , 11^0 , $(2/3)^0$, 4^{-3} , 5^{-3} , 6^{-2} , $(1/3)^{-3}$, $(3/5)^{-2}$, $(0.04)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q006 Written

Simplify the twelve same-base expressions in given Question 2: $a^2a^4, a^{-2}a^5, a^{-3}a^{-4}, a^3 \div a^5, a^8 \div a^{-3}, a^{-6} \div a^{-2}, (a^{-1})^3, (a^3b^4)^{-2}, (a^{-3}b^2)^{-3}, 3^53^{-3}, 4^44^{-3}, 5^55^{-5}$.

Classified Practice

Laws of Exponents and Roots

Q007 Written

Simplify given Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^23^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^42^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, **and** $(a/2)^3a^{-4} \div (2a)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q008 Written

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2} (4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2)$.

Q009 Written**Find $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.**

Classified Practice

Laws of Exponents and Roots

Q010 Written

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[6]{8})^2$, $\sqrt[3]{\sqrt{729}}$, and $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Q011 Written**Find** $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

Classified Practice

Laws of Exponents and Roots

Q012 Written

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, and $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

Q013 Written

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

Classified Practice

Laws of Exponents and Roots

Q014 **Written**

Perform the fifteen **given** calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

Classified Practice

Laws of Exponents and Roots

Q015 Written

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

Q016 Written

Express in the form a^r : $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

Classified Practice

Laws of Exponents and Roots

Q017 Written

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, and $[(16/9)^{-3/4}]^{2/3}$.

Q018 Written

Express in the form a^r : $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.

Classified Practice

Laws of Exponents and Roots

Q019 Written

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, and $[(27/125)^{1/2}]^{-4/3}$.

Q020 Written

Express the nine given expressions in the form a^r , including roots of a and numerical bases 2, 3, 5.

Classified Practice

Laws of Exponents and Roots

Q021 Written

Evaluate the twelve given expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, $[(1/25)^{1/3}]^{1/2}$, $[(1/8)^{1/2}]^{1/3}$, $[(1/27)^{1/3}]^{1/2}$

Classified Practice

Laws of Exponents and Roots

Q022 Written

Simplify $\left[\left(\left(1000/27\right)^{-2/3}\right)^{7/15}\right]^{-15/14}$.

Q023 Written

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$, (b) $6^{1/2}36^{1/4}$, (c) $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$, (d) $\sqrt{2} \div \sqrt[6]{32} \times \sqrt[3]{16}$.

Classified Practice

Laws of Exponents and Roots

Q024 Written

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7}\sqrt[6]{7}$, and $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

Q025 Written

For $a > 0, b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

Classified Practice

Laws of Exponents and Roots

Q026 Written

Perform the twelve numerical calculations in given Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

Q027 Written

For $a > 0, b > 0$, simplify the six **given** expressions built from $\sqrt{a}, \sqrt[3]{a}, \sqrt[4]{a}, \sqrt[5]{a}$ and mixed a,b radicals.

Classified Practice

Laws of Exponents and Roots

Q028 Written

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}} \right)^{-3/2}$.

Q029 Written

Assess the statement $\sqrt{a^2} = a$ for all real a , using positive and negative examples.

Classified Practice

Laws of Exponents and Roots

Q030 MCQ

The value of 2^{-3} is:**Choose the correct option.**

A 8

B $1/8$ C -8 D $1/6$

Q031 MCQ**When multiplying equal bases, we:****Choose the correct option.**

- A subtract exponents
- B add exponents
- C multiply bases only
- D take reciprocals

Classified Practice

Laws of Exponents and Roots

Q032 MCQ

The expression $a^5 \div a^{-3}$ simplifies to:

Choose the correct option.

A a^2

B a^8

C a^{-8}

D a^{15}

Q033 MCQ**A negative exponent means:****Choose the correct option.**

- A multiply by -1
- B take the reciprocal power
- C set the base to zero
- D add one

Classified Practice

Laws of Exponents and Roots

Q034 MCQ

The value of $(0.04)^{-2}$ is:**Choose the correct option.**

- A 25
- B 400
- C 625
- D 1600

Q035 MCQ**The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:****Choose the correct option.**

A -2

B -1

C 2

D 6

Classified Practice

Laws of Exponents and Roots

Q036 MCQ

The positive-exponent form of the 6-1 challenge is:**Choose the correct option.**

A a^7b^6

B $1/(a^7b^6)$

C $a^{-7}b^{-6}$

D $1/(a^6b^7)$

Q037 MCQ**The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:****Choose the correct option.****A** $1/3$ **B** 1**C** 3**D** 9

Classified Practice

Laws of Exponents and Roots

Q038 MCQ

 $\sqrt[3]{64}$ equals:

Choose the correct option.

A 2

B 4

C 8

D 16

Q039 MCQ $\sqrt[3]{12}\sqrt[3]{18}$ equals:**Choose the correct option.**

A 4

B 5

C 6

D 9

Classified Practice

Laws of Exponents and Roots

Q040 MCQ

The fourth root of $1/16$ is:**Choose the correct option.**A $1/2$ B $1/4$

C 2

D 4

Q041 MCQ

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:

Choose the correct option.

- A $\sqrt[3]{2}$
 - B $2\sqrt[3]{2}$
 - C $3\sqrt[3]{2}$
 - D 0
-
-
-

Classified Practice

Laws of Exponents and Roots

Q042 MCQ

 $\sqrt[4]{625}$ equals:

Choose the correct option.

A 4

B 5

C 5^2

D 25

Q043 MCQ**When combining roots with the same index, multiply:****Choose the correct option.**

- A the indices
- B the radicands
- C the exponents only
- D the signs only

Classified Practice

Laws of Exponents and Roots

Q044 MCQ

The roots challenge evaluates to:**Choose the correct option.**

A 0

B 1

C 2

D 4

Q045 MCQ

$\sqrt[5]{2^4}$ in rational-index form is:

Choose the correct option.

A $2^{4/5}$

B $2^{5/4}$

C 2^9

D $4^{1/5}$

Classified Practice

Laws of Exponents and Roots

Q046 MCQ

The value of $27^{2/3}$ is:**Choose the correct option.**

A 3

B 6

C 9

D 18

Q047 MCQ**A reciprocal square root is written as:****Choose the correct option.**

A $a^{1/2}$

B $a^{-1/2}$

C a^{-2}

D a^2

Classified Practice

Laws of Exponents and Roots

Q048 MCQ

The value of $81^{-3/4}$ is:**Choose the correct option.**

A 27

B $1/9$ C $1/27$ D $1/81$

Q049 MCQ

The exponent of a in $\sqrt[4]{a^3}$ is:

Choose the correct option.

- A $3/4$
- B $4/3$
- C $-3/4$
- D 3

Classified Practice

Laws of Exponents and Roots

Q050 MCQ

The value of $[(64/27)^{-4/3}]^{-1/2}$ is:

Choose the correct option.

A $4/9$

B $9/4$

C $16/9$

D $27/64$

Q051 MCQ**The 6^{-3} challenge simplifies to:****Choose the correct option.**A $3/10$ B $10/3$ C $100/9$ D $9/100$

Classified Practice

Laws of Exponents and Roots

Q052 MCQ

 $2^{2/3} 2^{1/2} \div 2^{7/6}$ equals:

Choose the correct option.

A $1/2$

B 1

C 2

D 4

Q053 MCQ $4^{5/6} 2^{4/3}$ equals:**Choose the correct option.****A** 4**B** 6**C** 8**D** 16

Classified Practice

Laws of Exponents and Roots

Q054 MCQ

 $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ equals:

Choose the correct option.

A $a^{1/3}$ B $a^{2/3}$ C $a^{5/3}$

D a

Q055 MCQ**The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:****Choose the correct option.**

A 2

B 3

C 6

D 12

Classified Practice

Laws of Exponents and Roots

Q056 MCQ

The given expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

Choose the correct option.

- A integer powers
- B combining rational exponents
- C logarithms
- D trigonometry

Q057 MCQ**The printed decimal in the 6-4 challenge is:****Choose the correct option.**

- A 0.04
- B 0.004
- C 0.4
- D 4

Classified Practice

Laws of Exponents and Roots

Q058 MCQ

For real a , $\sqrt{a^2}$ equals:

Choose the correct option.

- A a always
- B $-a$ always
- C $|a|$
- D 0

Quiz MCQ 1 **Concept Quiz · MCQ**

Which law gives $a^m a^n = a^{m+n}$?

Choose the correct option.

- A same-base multiplication
- B power of a power
- C reciprocal law
- D root product

Classified Practice

Laws of Exponents and Roots

Quiz MCQ 2 **Concept Quiz · MCQ**

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

Choose the correct option.

- A 2
- B 5
- C 10
- D 125

Quiz MCQ 3 Concept Quiz · MCQ

The value of $64^{-3/2}$ is:

Choose the correct option.

A 512

B $1/64$

C $1/512$

D 64

Classified Practice

Laws of Exponents and Roots

Quiz MCQ 4 Concept Quiz · MCQ

For a negative real a , $\sqrt{a^2}$ is:

Choose the correct option.

A a

B $-a$

C 0

D a^2

Quiz WRITTEN 5 Concept Quiz · Written**Simplify** $a^{-3}a^7 \div a^2$.

Classified Practice

Laws of Exponents and Roots

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Quiz WRITTEN 7 Concept Quiz · Written**Evaluate** $125^{-2/3}$.

Classified Practice

Laws of Exponents and Roots

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $\sqrt{a^2} = a$ is not true for every real a .

Answer key

Use the question number shown on each page.

Q001 $1, \frac{1}{8}, 25$

Q002 $a^3, a^8, \frac{b^2}{a^2}, 32$

Q003 $1, \frac{1}{9}, 16$

Q004 $a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$

Q005 $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Q006 $a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 4$

Q007 $25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}$

Q008 $\frac{1}{a^7b^6}$

Q009 $4, \frac{1}{3}$

Q010 $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 $3, \frac{1}{2}$

Q012 $4, 2, 10, \sqrt[3]{4}, 4\sqrt[3]{3}$

Q013 $6, 5, 3, \frac{1}{4}, \frac{1}{5}, \frac{1}{3}, 1, 10, 0.1$

Q014 $2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Q015 1

Q016 $5^{1/2}, 2^{4/5}, 3^{-3/5}$

Q017 $9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$

Q018 $a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q019 $27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$

Q020 $2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q021 $125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q022 $\frac{10}{3}$

Q023 1, 6, 6, 2

Q024 5, 8, 7, 9

Q025 $a^{1/3}, b\sqrt{a}$

Q026 $1, 9, 5, 9, \sqrt{5}, 4, 5, 3, 2, 1, 8, \sqrt[3]{3}$

Q027 $a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$

Q028 $\left(\frac{2025\sqrt{10}}{64}\right)^{3/2}$

Q029 Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Q030 B

Q031 B

Q032 B

Q033 B

Q034 C

Q035 A

Q036 B

Q037 B

Q038 B

Q039 C

Q040 A

Q041 B

Q042 B

Q043 B

Q044 B

Q045 A

Q046 C

Q047 B

Q048 C

Q049 A

Q050 C

Q051 B

Q052 B

Q053 C

Q054 A

Q055 C

Q056 B

Q057 B

Q058 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 a^2

Quiz WRITTEN 6 $2\sqrt[3]{2}$

Quiz WRITTEN 7 $1/25$

Quiz WRITTEN 8 $\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$